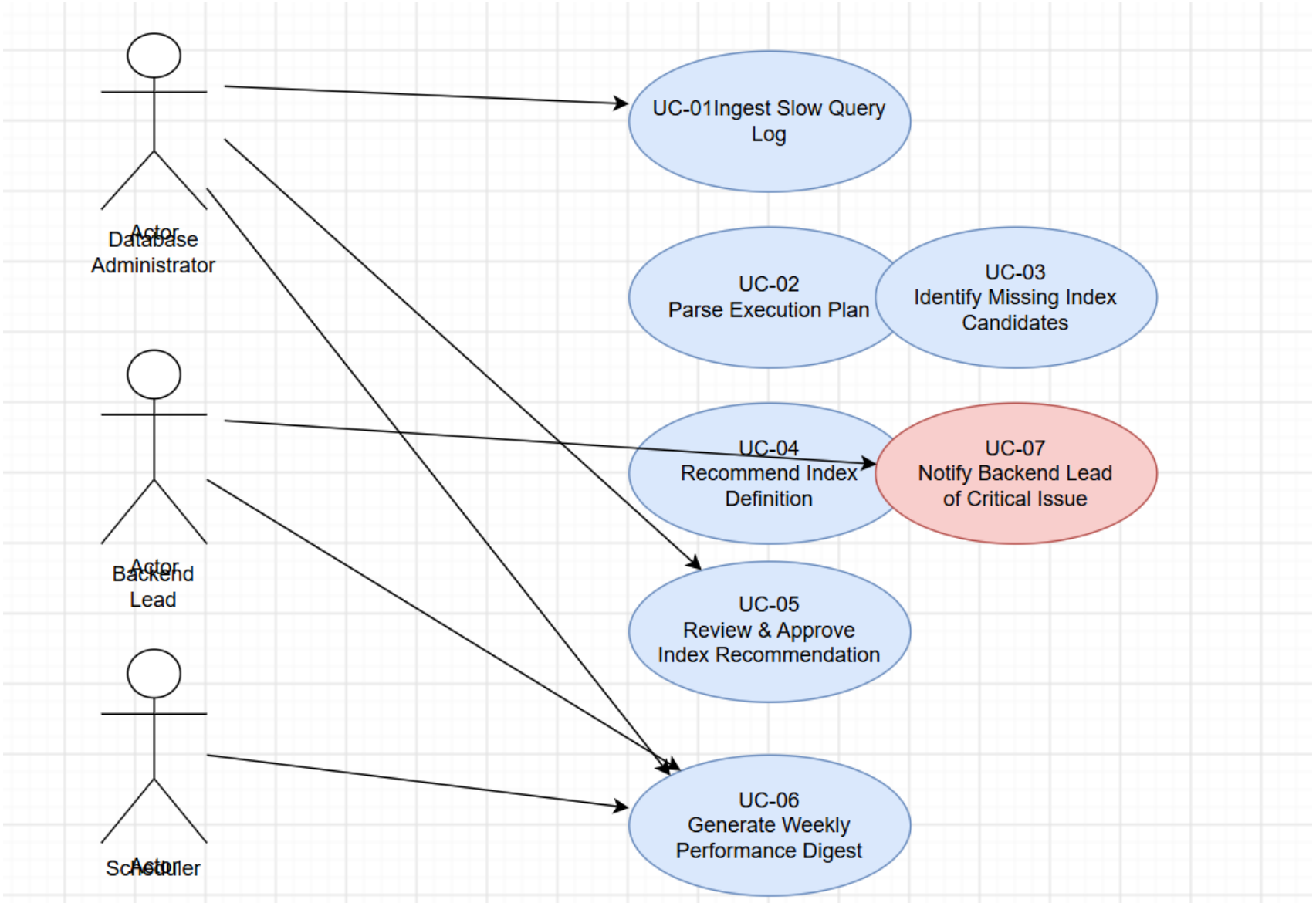
NAME : S V BHAVANA
SRN : PES2UG24CS417
SECTION : G

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse Postgres/MySQL EXPLAIN plans, highlight sequential table scans on large tables, and recommend optimal index column definitions.	High	Pass: Missing composite index identified on filtered columns. Fail: Query plan with sequential scan marked optimized.	Core capability that enables all downstream index recommendations.
FR-002	Functional	The system shall ingest slow query log files/streams from Postgres and MySQL sources and normalize them into a common internal schema.	High	Pass: A 10,000-record sample log file is ingested and 100% of records appear in the normalized store within 60 seconds. Fail: any record is lost, duplicated, or ingestion exceeds 60 seconds.	Reliable ingestion is the entry point of the pipeline; without it no downstream analysis is possible.
FR-003	Functional	The system shall estimate the performance impact (e.g., reduction in scan cost or execution time) of each recommended index before presenting it to the DBA.	Medium	Pass: Each recommendation displays an estimated cost reduction percentage computed from the query planner's cost model. Fail: recommendation shown without a quantified impact estimate.	Helps the DBA prioritize which indexes to implement first when multiple candidates exist.
FR-004	Functional	The system shall allow a Database Administrator to review each recommended index and mark it as Approved or Rejected.	High	Pass: DBA changes a recommendation's status via the dashboard/API and the new status is persisted and reflected within 2 seconds. Fail: status change is not saved or is not visible after refresh.	Keeps a human decision-maker in control of schema changes, since applying an index is not fully automatic.
FR-005	Functional	The system shall automatically generate a weekly digest summarizing the top slow queries, missing index candidates, and their estimated impact, and make it available to the Backend Lead.	Medium	Pass: A digest is generated every Monday at 06:00 containing all queries with average execution time above 500ms from the prior 7 days. Fail: digest is missing, incomplete, or delayed by more than 1 hour.	Gives the Backend Lead a recurring, low-effort summary instead of requiring manual log review.
NFR-001	Non-Functional (Performance & Security)	Slow query log ingestion must handle up to 5,000 log records per minute without dropping entries or slowing the host DB.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: entries dropped or host DB latency increases beyond agreed threshold.	Guarantees the profiler does not itself become a performance or security risk to production databases.
NFR-002	Non-Functional (Reliability & Availability)	The performance dashboard and weekly digest must be available with at least 99.5% uptime, and dashboard pages must load in under 2 seconds under normal load.	Medium	Pass: Load testing over a 7-day period shows uptime at or above 99.5% and average page load under 2 seconds. Fail: uptime falls below 99.5% or average page load exceeds 2 seconds.	DBAs and Backend Leads rely on timely access to recommendations; a slow or unavailable tool defeats its purpose.

2.UML Use-Case Diagram



3.Use-Case Flow Document

Main Success Scenario

1. System retrieves a parsed execution plan flagged with a sequential scan on a large table.
2. System analyzes the filtered and joined columns referenced by the query to determine the optimal index candidate.
3. System calculates the estimated performance improvement (e.g., percentage reduction in scan cost) from applying the candidate index.
4. System generates a concrete index definition, including target table, column order, and index type (e.g., single-column or composite B-tree).
5. System stores the recommendation with status "Pending Review" and links it to the originating query and its cost estimate.
6. System displays the recommendation, with its estimated impact, on the Database Administrator's dashboard.
7. Use case ends successfully.

Alternate Flow

3a. Critical Severity Detected

a) System determines the query's estimated cost exceeds the configured critical threshold.
b) System marks the recommendation's severity as "Critical".
c) System triggers the extension use case UC-07 (Notify Backend Lead of Critical Issue), sending an alert containing the query details and recommendation summary.
Flow resumes at step 4 of the Main Success Scenario